Our Gantry System:

The CSUN ARCS sulfur concrete printer is a research project exploring whether Martian habitats can be 3D printed using sulfur-based concrete, a material that can be made from resources available on Mars. Our project, by contrast, is a gantry-based material handling system designed to move and lift objects across a defined workspace. Both are large-format gantry systems that rely on software to control movement, and both must handle heavy, difficult materials. The main difference is the task: the concrete printer deposits material layer by layer to build a structure, while our system picks up, carries, and lowers discrete objects.

For routing and movement, the team assembled and modified a large-format gantry 3D printer running Marlin firmware, which moves the nozzle along set paths to lay down each layer. Because the extruder height was fixed, they built a mechanical lowering system using synchronized car jacks driven by stepper motors to handle vertical motion. This is relevant to us because it shows stepper-controlled lifting of a heavy load, similar in concept to our hook's lifting and lowering. We can follow the idea of using established, open-source motion firmware to handle coordinate movement and the idea of keeping multiple lifting points synchronized. However, the project page does not describe their homing, position feedback, or motor sizing, so we will still need to determine those for our own system. Our system also differs in that it moves to discrete target locations rather than following continuous print paths, and we plan to simulate movement and routing before building the prototype.

For safety, the project focuses on reducing risk through automation, aiming to automate key steps so less human intervention is needed in hazardous Martian conditions. Their other safety-related work involves the material itself, since the move to sulfur requires a heated pump system and components that can withstand high temperatures. The page does not mention an emergency stop, movement limits, load limits, or fault detection. This is a significant gap for our purposes, because our core safety features include an emergency stop, enforced movement limits, and a load-limit check for a suspended load. We can follow the general principle of automating operations to keep people away from hazards, but our machine safety features will need to be designed by our team or drawn from other sources, such as overhead crane or hoist safety standards.

For attachments, the printer uses a single tool: a custom auger-based concrete pump integrated with an air compressor. The team tested flow problems related to air-tightness, tube bends, and system pressurization, and concluded that industrial rotor-stator pumps might deliver better pressure but are currently too expensive. There is no interchangeable tool system, so this project does not address our long-term goal of swappable attachments. What we can take from it is its iterative, test-driven approach to a custom end effector: building a working version, identifying weaknesses through repeated trials, and documenting results. That approach fits our Version 0 plan of starting with a single hook attachment and refining it before adding other tools.

Overall, the concrete printer covers gantry movement well and offers a useful example of heavy, stepper-driven vertical motion through its synchronized jack system. It covers safety only at a

general level and does not address interchangeable attachments. Its biggest differences from our project are its purpose, building structures from extruded material rather than moving objects, and its lack of documented machine safety features such as an emergency stop or load limits. It is most useful to us as a reference for large-format gantry construction and for lifting heavy loads with synchronized motors.

Sources:

- 3D Additive Manufacturing of Sulfur Concrete for Mars Habitats – CSUN ARCS

Concrete Printer: